SPACE DEVELOPMENT AGENCY

Proliferated Warfighter Space Architecture

Tranche 3 Tracking Layer

Program Solicitation SDA-PS-25-02

31 January 2025

Draft

Table of Contents

1. Introduction

The Space Development Agency (SDA) is issuing this Other Transaction Authority (OT) solicitation for a prototyping effort of the Proliferated Warfighter Space Architecture (PWSA) Tranche 3 (T3) Tracking Layer. The T3 Tracking Layer prototyping effort will accelerate the capability to provide global, persistent indications, detection, warning, tracking, and identification of conventional and advanced missile threats, including hypersonic missile systems. The T3 Tracking Layer will also demonstrate a missile defense capability. The T3 Tracking Layer scope includes the procurement and deployment of an anticipated minimum of 54 space vehicles (SVs) with infrared (IR) sensors in six (6) orbital planes. Additional prototyping of satellites and sensor payloads may also occur under this OT in order to inform requirements and constellation design. The fully deployed Tracking Layer is estimated to include more than 100 SVs in low Earth orbit (LEO) across multiple planes.

All references in this solicitation to Attachment 1 – Statement of Work (SOW) and Attachment 3 – Technical Requirements Document (TRD) pertain to the T3 Tracking Layer Statement of Work and Technical Requirements Document inclusive of its classified appendix.

1.1 Motivation

SDA has determined that the proliferation of various mission functions across the PWSA using commodity SVs is the most efficient and cost-effective means of evolving the Architecture. To accomplish this, SDA employs a capability-focused business model prioritizing speed and lowering costs by harnessing commercial development to achieve proliferation and enhance resilience. The objective is to establish and demonstrate the ability to rapidly evolve development processes and field capability on a significantly faster timeline than historical space systems acquisition. In pursuit of this objective, SDA requires each PWSA Performer's SVs and communications systems to be interoperable with the SVs and systems developed by all other PWSA performers as well as those to be deployed in Tranche 1. Additionally, all SVs must operate in an integrated fashion through a common ground system.

1.2 Program Approach

This solicitation seeks Other Transaction (OT) proposals under the authority of 10 U.S.C. § 4022 for U.S. space industry builder-operator teams that will engage with SDA to develop and operate the T3 Tracking Layer as part of the PWSA. The Government is purchasing a nominal constellation of 54 (or more) satellites that are sufficient to meet constellation coverage and performance objectives for the T3 Tracking Layer. SDA anticipates purchasing these SVs divided into six (6) orbital planes, to be awarded to more than one (1) vendor, subject to available funding.

The T3 Tracking Layer constellation will include a mix of Missile Warning/Missile Tracking (MW/MT) and Missile Warning, Tracking and Defense (MWTD) infrared (IR) mission payloads. Three awards will be made for 18 SVs with either MW/MT or MWTD capability. Each Offeror may propose up to two submissions; one to develop and deliver at least 18 MW/MT and/or one to develop and deliver 18 MWTD SVs. Each submission shall deliver 18 satellites in two (2) orbital planes, with the associated ground support and operations and sustainment (O&S) capability. Each of the planes may be populated with SVs containing single sensor to multiple sensor payloads supporting the constellation coverage objectives so long as each plane may be populated with a single launch on the NSSL contract.

Note that for SDA to make an OT award to the Offeror, at least one of the following conditions must be met:

1. There is at least one (1) nontraditional defense contractor or nonprofit research institution participating to a significant extent in the prototype project.
2. All significant participants in the transaction other than the Federal Government are small businesses (including small businesses participating in a program described under section 9 of the Small Business Act (15 U.S.C. 638)) or nontraditional defense contractors.
3. At least one-third of the total cost of the prototype project is to be paid out of funds provided by sources other than the Federal Government.
4. The senior procurement executive for the agency determines in writing that exceptional circumstances justify the use of a transaction that provides for innovative business arrangements or structures that would not be feasible or appropriate under a contract or would provide an opportunity to expand the defense supply base in a manner that would not be practical or feasible under a contract.

1.3 Program Plan and Schedule

The first plane of the T3 Tracking Layer constellation will be launched no later than **31 March 2029** with the launches of subsequent planes expected to follow on one-month intervals. The Offeror is expected to plan their program in accordance with the milestone dates and deliverables specified in Attachment 1 – SOW and Attachment 3 – TRD of this solicitation.

1.4 T3 Transport and T3 Tracking Concept of Operations (CONOPS) Overview

The Tranche 3 Tracking Layer establishes initial launch capability (ILC) in 31 March 2029 and continues with an approximately six-month-long, monthly launch campaign. As these SVs are placed into their insertion orbits, SDA will undergo a continuous checkout and commissioning process to prepare Tranche 3 for acceptance into operations conducted out of the Operations Centers (OCs) at Grand Forks, AFB (OC-N) and Redstone Arsenal, AL (OC-S).

After being placed into an insertion orbit by an NSSL launch vehicle, each plane of SVs will undergo LEOPs, which includes the Separation / Initialization / Checkout / Orbit Raising (SICO) phase and the Functional Acceptance Test (FAT).

SICO includes initial acquisition of the SVs, SV and payload initialization and aliveness checks, orbit raising and phasing to the operational orbit, and Offeror preparation for Government- and Ground Segment (GS) integrator-led acceptance testing.

FAT is then conducted to ensure that each system delivered by the Offeror – comprised of a plane of SVs integrated with the Offeror's Network Established Beyond the Upper Limits of the Atmosphere (NEBULA) Operations – Vendor Architecture (NOVA) ground system – is fully mission capable. Once FAT and the subsequent Functional Acceptance Review (FAR) are successfully completed, each plane of Tranche 3 SVs will be transitioned to the Operations phase, with operations conducted from the OCs.

T3 commissioning activities will be conducted by the Offeror in coordination with the Tranche 3 Program Integrator, and these activities will be conducted from the Offeror's facilities in T3.

SUPERNOVA instances serving LEOPs may exercise a subset of the operational SUPERNOVA's functional capabilities but will replicate the SUPERNOVA-NOVA interface and key capabilities required to assess NOVA and SV functional capability prior to transition to Operations.

SDA envisions the "exit state" of the commissioning phase to be very similar to the "entry state" to Operations – a fully-functional, optically-interconnected, and networked plane of SVs commanded and controlled by Offeror personnel using their NOVA ground system with verified interfaces and connectivity with SUPERNOVA.

Once commissioning is complete, each plane of SVs will undergo a similar set of functional tests from the OCs during a brief Operational Acceptance Test (OAT) and begin their transition to Mission Operations. The Offeror will continue to support their vehicles for a period of up to five (5) years of operations following the transition to Mission Operations plus the time necessary for disposal.

2 System Description

The T3 Tracking Layer is a pLEO constellation at an altitude of about 1000 km, consisting of six (6) orbital planes with nine (9) SVs in each plane. Notional constellation parameters are provided in Table 2-1. Each SV is equipped with an IR mission payload, Optical Communications Terminals (OCTs), and K_a-band and S-band communications payloads.

The T3 Tracking Layer constellation will include a mix of MW/MT and MWTD infrared (IR) mission payloads. The intent is to achieve near-continuous global stereoscopic coverage for Missile Warning and Missile Tracking and along with SVs equipped with an IR mission payload capable of generating fire control quality tracks for Missile Defense.

This system includes on-orbit calibration capability, on-board processing of Tracking data, and minimization of solar outages. Each SV will have multiple pointing modes to include Nadir- and Ram-Fixed, Off-Earth Inertial Point/Track, Sun Pointing, Solar Avoidance Pointing, and Safe. The SVs must be able to conduct continuous operations without power or thermal constraints.

The T3 Tracking Layer SVs will include OCTs to support crosslinks in-plane, cross-plane, and to the Transport Layer, as well as space-to-ground optical links. Every T3 Tracking Layer SV will be interoperable with all PWSA SVs, and all other T3 Tracking Layer SVs, regardless of vendor.

Table 2-1 Notional T3 Tracking Layer Constellation

Parameter	T3 Tracking SVs	
	Mid-inclined	High-inclined
Number of Planes	3	3
SVs per plane	9	9
Altitude [km]	1020	1000
Eccentricity [-]	0	0
Inclination [deg]	~45	~81
Initial Anomaly [deg]	1.6, 16.6, 31.6	-1.5, -1.5, 14.08
Initial RAAN [deg]	0, 120, 240	21.693, 105.59, 155.9

The system description and requirements can be found in Attachment 3 – TRD of this solicitation. The Offeror shall propose a solution that meets the requirements as laid out in that attachment.

3 Selection Process

Within this solicitation, the following definitions are used:

- **Prime:** The Offeror
- **Major Teammate:** A member of the Prime's development team with a different CAGE code from the prime Offeror, that either accounts for no less than five (5) percent of the proposed cost and/or develops one (1) or more of the following subsystems or capabilities:
  - IR Sensor(s)
  - Onboard IR Mission Data Processor
  - Optical Communications Terminals
  - Space Vehicle Bus
  - System Integration
  - Operations and Sustainment People, Processes or Tools
  - NEBULA Operations – Vendor Architecture (NOVA) System
  - NOVA Ground Infrared Mission Data Processing

If the following subsystems are not procured as commercial off-the-shelf systems (COTS) and/or require significant modifications, then the developer of the subsystem shall be considered a Major Teammate:

- Networking and Data Routing
- Encryption
- K_a Mission Payload
- Backup TT&C System

Major Teammate proposals (regardless of teammate tier) shall contain the information requested in Section 0 to allow the Government to review the proposed price and supporting rationale. Major Teammates shall not submit Governance, Technical (or other) volumes. Note that certain information regarding Major Teammates is requested elsewhere in the solicitation but this information should be included as part of the Prime's proposal.

Major Teammate proposals shall follow the same font, spacing, table, and margin formatting instructions that apply to the Prime.

- **Commercial off-the-shelf:** COTS products are defined as software or hardware products that are commercially ready-made and available for sale, lease, or license to the general public. For the purposes of this solicitation, products that are less than TRL 7 cannot be considered COTS. **If a product requires modifications as part of the T3 Tracking program, then it cannot be considered COTS**.

4 Proposal Preparation Instructions

The Offeror shall submit their Tranche 3 Tracking Layer proposal via six (6) volumes. Specific volume contents and associated page limits are detailed below. The Offeror should feel free to arrange the order of the information within each volume as they see fit for clarity and, where page limitations apply, focus on the most critical information. In addition, the Offeror should focus on demonstrating the depth of their design and demonstrating through analysis that it achieves the necessary performance, rather than echoing all Government requirements in their response. Volumes I and V can be common for MW/MT and MWTD submissions, but Volume II, III, IV and VI must be specific to MW/MT or MWTD submissions.

Volume/Tab	Title	Page Limit	Due at 1200 EST on date specified below
Volume I	Governance	15 pages	21 May 2025
	Table of Contents/Cross Reference Matrix		
Tab 1	Executive Summary		
Tab 2	Program Management		
Tab 3	Security Performance		
Tab 4	Facilities Management		
Tab 5	Systems Engineering		
Tab 6	Mission Assurance		
Tab 7	Organizational Conflict of Interest (OCI)	Excluded from page count	
Tab 7	Key Personnel Résumés	Excluded from page count	
Tab 8	Personnel Matrix	Excluded from page count	
Tab 9	Technical Data and Computer Software Rights Assertion	Excluded from page count	
Volume II	Technical	50 Pages (Excludes Appendices)	21 May 2025
	Table of Contents/Matrix		
Tab 1	Executive Summary		
Tab 2	CONOPS Overview		
Tab 3	Space Vehicle Overview		
	Tranche 3 Tracking Layer		

	Payload Architecture and Design		
	Infrared Mission Payloads		
	Optical Communications Terminal (OCT)		
	Ka-band Mission Payload		
	Telemetry, Tracking and Control (TT&C)		
	Network and Data Routing		
	Navigation		
Tab 4	Space Vehicle Bus Design		
	Bus Design and Heritage		
	S-band Backup TT&C System		
	Directed Energy Resiliency System		
	Cybersecurity		
	Encryption		
Tab 5	Manufacturability		
	Supply chain		
Tab 6	Test, Demonstration and Verification		
Tab 7	Launch Services Overview		
Tab 8	Ground Segment Integration and Operations		
	FlatSat(s) and SoftSat(s)		
Tab 9	Mission Design		
Tab 10	Appendices		
	Radiometry Analysis	5 Pages	
	Metric Analysis	5 pages	
	Launch, Early Operations, Commissioning, and Transition to Operations Plan	5 pages	

	Parameter Specification	Excel	
	Technical Requirements Verification Methods		
Volume III	Integrated Master Schedule	10 Pages (Excludes Appendix)	21 May 2025
Tab 1	IMS		
Tab 2	Schedule Risk		
Tab 3	Appendix – Portfolio IMS	5 pages	
Volume IV	Price and Rationale	Unlimited	21 May 2025
Tab 1	Executive Summary and Pricing Rationale		
Tab 2	Price Proposal Structure		
	Bidding Unit		
	CLIN Structure		
	Work Breakdown Structure (WBS)		
	Milestone Payments		
	Delivery Incentive		
	On Orbit Penalty Performance		
	Milestone Payment Structure		
Tab 3	Proposal Pricing Details		
	Additional Pricing Data		
Tab 4	Govt Furnished Equipment, Information & Property		
Tab 5	Major Teammate Proposals		
Tab 6	Organizational Conflict of Interest (OCI)		
Volume V	Past Performance		21 May 2025
Tab 1	Executive Summary	2 pages	

Tab 2	Recent and Relevant Past Performance		
	Citations (up to 4) to include POC info	4 pages each (NTE 16 pages)	
Volume VI	Draft Other Transaction Agreement (OTA)	Unlimited	21 May 2025
Note: Page numbering shall re-start at the beginning of each section or volume. Note: Classified Addendums are included in the page count identified above. Note: Cover pages, tables of contents, appendices and acronym tables are not part of the page count in any volume and are optional for Volumes III-VI			

- **Volume I – Governance** shall describe how the Offeror's proposed program management, systems and mission engineering approaches, processes, and tools enable the Offeror to execute the program on schedule and within the price proposed. This volume shall also contain résumés for key personnel who will perform the T3 Tracking Layer program and a complete listing of data rights asserted by the Offeror and all of its subcontractors, teammates, and/or vendors.
- **Volume II – Technical** shall contain a detailed description of the Offeror's proposed technical solution to accomplish the requirements defined in Attachment 1 – SOW and Attachment 3 – TRD.
- **Volume III – Schedule** shall contain a complete Integrated Master Schedule. This volume shall also provide: an analysis of schedule risks; identification of key milestones; entrance criteria, key accomplishments and exit criteria for significant reviews and activities; and a clear presentation and analysis of the critical paths(s) from contract award through SV delivery.
- **Volume IV – Price and Rationale** shall contain the Offeror's Firm Fixed Price and associated detail.
- **Volume V – Experience and Past Performance** should contain at least one (1) and may contain up to four (4) recent (within the last five (5) years), relevant examples of delivery of complex systems, subsystems, or components, preferably space- related hardware or software. Examples must include the offeror's schedule and cost/price results for each. Offerors may also include at least one (1) and up to two (2) past performance examples for each Major Teammate.
- **Volume VI – Draft Other Transaction (OT) Agreement** shall contain the Offeror's proposed T3 Tracking Other Transaction Agreement, including all attachments.

Normal text for all volumes shall be in Times New Roman font, and the font size shall be no smaller than 12 pt. All text in figures and tables for all volumes shall be in Arial Font, and the font shall be no smaller than 10 pt. A table shall consist of at least two (2) columns in every row. Tables or figures that are deemed attempts to circumvent these text formatting restrictions will be removed from the proposal and not evaluated. All margins (top, bottom, left, and right) shall be no smaller than one (1) inch. Single line spacing shall be used for all text in paragraphs, bulleted lists, and tables. All pdf pages, in all volumes, formatted larger than 8.5 inches by 11 inches shall be counted as two (2) pages. All pdf pages, in all volumes, formatted larger than 11 inches by 17 inches will be removed from the proposal and not evaluated. Line numbers shall be applied to all text sections in all volumes and shall be set within the left margin of the page so as not to reduce the usable size of the page. Title pages and tables of contents, tables of figures, and tables of tables will not count against the page counts of the respective volumes and will not be evaluated for content. **Note: Nonconforming proposals may be rejected without review.**

- A Table of Contents shall be incorporated into Volume I & II and will include a listing of the section titles and subsection titles. Page numbers for each title must be indicated. The Table of Contents shall also include a listing of illustrations and tables with page numbers. A cross-reference matrix for Volume I & II is also required. This matrix shall indicate the proposal volume section, subsection, and paragraph, cross-referenced with the appropriate SOW paragraphs, applicable solicitation instructions; evaluation criteria, and deliverables/CDRLs, as applicable. A glossary of acronyms and terms for Volume I & II is required. The use of acronyms without definitions within the proposal volume is acceptable; however, the Offeror shall ensure that all acronyms are defined in the glossary for that volume. These pages will not be included in the page count for any volume.

4.1 Volume I – Governance

Volume I shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable **content, other than those specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated.**

4.1.1 Executive Summary

The Offeror shall present their proposed system governance solution for the T3 Tracking Layer, including their team construct and expected contributions by each Major Teammate. The Executive Summary should include a summary of the Offeror's programmatic and systems engineering approaches, to include the cost and risk management approaches.

4.1.2 Program Management

The Offeror shall demonstrate their ability to manage and conduct the program to a successful completion on time and within the agreed upon price. The Offeror shall discuss their program, personnel, cost, and risk management approaches, including any processes and tools. The Offeror shall include a list of the primary technical and schedule risks on the project, the planned mitigations, and the approach for managing risks throughout the system life cycle. Risks and mitigations should be focused on execution of the proposed effort. Decisions made while developing the proposed solution (e.g., investments made prior to proposal submission, etc.) are appropriate to cite as risks or mitigations. Schedule management shall be discussed in Volume III; however, the Offeror is free to cross-reference Volumes I and III if doing so will be of benefit in explaining the value and quality of their approach. The Offeror should describe processes to staff the program with qualified personnel, identify and manage critical vendors and subcontractors, and control costs within a FFP environment.

The Offeror shall describe the schedule and performance risk associated with other commitments by the Prime and subcontractor teams and facilities during the period of performance, including, but not limited to, commitments to other SDA programs. If the Offeror supports other SDA programs, the Offeror shall clearly define how they will support T3 Tracking with minimal disruptions to their existing SDA programs, inclusive of personnel and facilities.

The Offeror shall describe their approach to retaining or obtaining appropriately cleared staff and facilities to support classified work.

The Offeror shall document planned communications with the Government, detailing the purpose, entrance criteria, key accomplishments anticipated, exit criteria, and documentation methodologies for each meeting, review, or key management milestone within the program.

The Offeror shall plan the program using the program reviews identified in Section 2.1 of Attachment 1 – SOW and in accordance with other dates specified throughout Attachment 1. Table 4-1The Offeror shall explicitly state the need for any anticipated waivers to Government requirements, standards, and/or compliance documents specified in Attachment 1 – SOW and Attachment 3 – TRD and its classified appendix. Waivers shall describe how the Offeror's approach will meet mission needs and provide supporting trades analysis.

4.1.2.1 Subcontractor and Supply Chain Management

The Offeror shall address their subcontractor and supply chain management methodologies. The Offeror shall include a table showing the delivery schedule of critical parts. The Offeror shall present a strategy for managing delivery performance of major subcontractors with finite production capacity to accomplish this program.

The Offeror should describe how they will minimize the effect on the program from fluctuations in supply chain availability and subcontractor deliveries. The Offeror should propose a contingency plan in response to supply shortages for all critical parts to include pricing, supply, and delivery information. The Offeror shall include a sparing plan for critical and long lead parts/units addressing acquisition of critical spares, inventory sources, and associated risks.

The Offeror should provide an assessment of hardware, firmware, and software maintainability, including the availability of manufacturer/developer support for all elements of the design to ensure that enduring product support for security patches and maintenance support remains available over the operational life of the system.

4.1.2.2 Nontraditional Defense Contractors (NDC) / Nonprofit Research Institution

If applicable, the Offeror shall identify which of their subcontractors constitutes an NDC or a nonprofit research institution in accordance with 10 U.S.C. 2302(9). The Offeror shall identify the extent of participation of NDC(s) and/or nonprofit research institution(s) to the Offeror's T3 Tracking Layer program.

4.1.3 Security Performance

The Offeror shall describe the Offeror's security approach and ability to comply with the security requirements of this solicitation. It should contain the Offeror's security approach, the approach to meeting the clearance requirements for proposed personnel, facility locations, and classified information safeguarding.

4.1.3.1 Facilities Clearance

The Offeror shall identify its proposed location(s) for performing classified efforts for this program. The Offeror is required to hold a current Top Secret facility clearance issued by DCSA in accordance with 32 CFR Part 117 and must provide the complete address and CAGE Code of the facility using Attachment – Personnel & Facilities Template.

4.1.3.2 Key Personnel Clearances

The Key Personnel listed in section 3.1.1.8 MUST have current, Top Secret (TS) clearances with SCI eligibility and be available to work at contract start. The Offeror shall provide the required information for the Key Personnel listed in section 3.1.1.8 using Attachment – Personnel & Facilities Template.

4.1.3.3 Foreign Ownership and Controlling Interest

The Offeror shall describe its approach for satisfying all requirements of this solicitation related to Foreign Ownership, Control, or Influence (FOCI). The Offeror's security approach should identify, at a minimum, the following:
Compliance with Disclosure of Ownership or Control by a Foreign Government (DFAR 252.209-7002) and shall include submission of a current SF 328, "Certificate Pertaining to Foreign Interests (FOCI)" if applicable. Offerors must provide the same information for each proposed subcontractor or subsidiary if applicable. If Offeror's are foreign owned, they must have an approved FOCI mitigation plan and/or a Cognizant Security Agency (CSA) Security Control Agreement (SCA), or Special Security Agreement (SSA). For Companies that will require access to Collateral, SCI, or COMSEC information, a National Interest Determination (NID) is required.

4.1.3.4 Contract Security Classification Specification (DD254)

The Offeror shall complete DD254 and submit it with the Security Volume. The DD254 should include the CAGE code and address of all locations where classified work is anticipated.

4.1.4 Facilities Management

The Offeror shall detail the secure facilities that will be used to execute the T3 Tracking program. The Offeror may reasonably assume that SDA will support co-use arrangements but should not assume that facilities can be newly certified on a schedule suitable for supporting this program. The Offeror shall include a table listing the primary facilities or rooms to be used. For each item, the Offeror shall provide the area, the maximum classification level, the certification status, the certifying organization, and networks and voice communications capabilities available for use. The Government expects that regular access to classified networks is important for execution of this program.

4.1.5 Systems Engineering

The Offeror shall demonstrate how their proposed systems engineering processes and methods will be used to deliver a fully compliant T3 Tracking system. The Offeror shall clearly identify their approach to systems engineering and detail the use of any processes and tools that will be used on the program. The Offeror shall identify any reference material, such as corporate command media or published standards, that are used as the basis for their systems engineering approach and discuss how such material will be tailored for the T3 Tracking program. The Government expects that the Offeror will utilize system engineering best practices.

4.1.6 Mission Assurance

The Offeror shall describe the mission assurance approach and Offeror's quality systems. The Offeror should include a discussion of relevant standards and certifications. In particular, the Government is interested in how the Offeror plans to tailor – if applicable – existing mission assurance processes and tools to meet the requirements of a proliferated LEO constellation designed for a 60-month operational mission life lifetime.

4.1.7 Organizational Conflict of Interest

The Government has determined there is potential organizational conflict of interest due to the nature of the anticipated separate solicitations and contracts for the entire Tranche 3 Program. The entire, fully integrated T3 program will likely include multiple T3 SV contractor(s)/performers (primes and subs) and T3 Ground contractor(s)/performers (primes and subs) who will interface with the T3TRK Performer. Therefore, if a T3TRK Performer intends to bid or perform on any other T3 contract/agreement, there may be an OCI. In addition, the Space Development Agency's OCI policy clearly states that SDA SETA must remain OCI free within SDA; therefore, a contractor cannot concurrently be an SDA SETA contractor and a T3TRK Performer.

As a part of the proposal, the Offeror shall provide the Agreements Officer with complete information of previous, potentially impending, or ongoing work that is in any way associated with this acquisition. The Offeror shall provide a written assessment addressing why there is no OCI and whether a future OCI is likely or will be avoided. In the event there is a known OCI or if the bidder intends to bid on future T3 Program acquisitions, the Offeror shall include an OCI Plan in addition to the Assessment. An Offeror who assesses they will avoid OCI with the T3 Program MUST include a statement (guarantee) attesting they will avoid OCI for the life of this T3TRK Agreement. If no OCI exists or will exist, the Offeror need only provide the Assessment. If a conflict exists or may exist, both the Assessment and the OCI are required to be a complete proposal.

4.1.8 The resulting Agreement will include an organizational conflict of interest limitation applicable to the related Government scope of work, at either a prime contract level, at any subcontract tier, or both. During evaluation of proposals, the Government may also, after engagement with the Offeror and consideration of ways to avoid the conflict of interest, insert a special provision addressing eligibility for future contract awards within the existing Tranche 3 Program. The OCI mitigation plan will not be included in the Volume I page limit. Key Personnel Résumés

The Offeror shall identify the key personnel planned to be assigned to this program from the Offeror and Major Teammates as appropriate. Résumés should be in bullet format and include:

- Summary statement highlighting experience and expertise

- Current role with the Offeror or teammate to include current assignments and programs with particular attention paid to efforts currently underway for SDA to include percentage of time already applied to other SDA programs or projects
- Planned role within the T3 Tracking Layer program
- Evidence of applicable mission experience or understanding
- Education (academic and military, if applicable)
- Anticipated percentage of time to be assigned to this program; if already dedicated to an SDA effort the Offeror should clearly indicate how current duties would be absorbed by alternate personnel (if applicable) so as to preclude any negative affects to ongoing efforts
- Signed statement agreeing to support the SDA T3 Tracking Layer program if awarded

Proposed staff members not currently employed by the Offeror may be included as appropriate. In those cases, the Offeror shall include signed contingent offer letters. These letters should be signed by both the Offeror and the proposed staff member and should state the willingness of the proposed staff member to join the Offeror in support of the SDA T3 Tracking Layer program if awarded.

At a minimum, key personnel shall include the staff supporting the proposed program in the following positions:

From Offeror:

- Program Manager
- Chief Engineer
- IR Mission Payload Lead
- Assembly, Integration, and Test (AI&T) Lead
- Subsystem Lead(s) (if Offeror is developing any of the major subsystems. The major subsystems are those listed in the Major Teammate definition in Section 3.)

Information Security Certified Professional

From each Major Teammate:

- Program Manager
- Chief Engineer

4.1.9 Personnel Matrix

The Offeror shall compare and contrast their submitted personnel résumés against the SDA T3 Tracking Layer requirements to validate corporate and individual experience and expertise thereby giving the Government confidence in the Offeror's probability of success. Format for this comparison is left to the Offeror discretion, but this section of the proposal should stand alone for evaluation. Any personnel currently working any other SDA program or project on behalf of the Offeror, or any Major Teammate shall be clearly identified, to include their percentage time dedicated across the Offeror or teammate's SDA portfolio.

4.1.10 Technical Data and Computer Software Rights Assertions

The Offeror shall clearly delineate **all** limitations on data, hardware, and computer software rights for technical data, hardware, or computer software provided by the Offeror and **all of its**

teammates, in the format shown in Attachment 4 of the Draft Other Transaction Agreement (OTA). It is the Government's assumption that ***no*** hardware, technical data, or computer software will be provided with less than Government Purpose Rights. and that the vast majority of information generated will be provided with Unlimited Rights to the Government. The Offeror shall clearly identify those elements to be provided with less than Unlimited Rights and identify the rationale for their assertion(s) using the format shown in Attachment 4. Any item related to the performance of this effort and not contained in Attachment 4, shall be delivered upon request to the Government, with no less than Government Purpose Rights. The terms "Unlimited Rights", "Government Purpose Rights", "Limited Rights", and "Restricted Rights" are defined in Volume VI Draft Other Transaction (OT) Agreement.

4.2 Volume II – Technical

Volume II shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable content, **other than those specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated**. With the exception of Commercial Data Sheets, Volume II appendices may also be classified.

4.2.1 Executive Summary

The Offeror shall present their proposed solution concept for two (2) planes of the T3 Tracking Layer and its NOVA ground system. The Executive Summary shall include an Operational Viewpoint (OV)-1 or equivalent high-level overview of the proposed solution. The graphics shall depict the SVs, their payloads, and the means by which they provide data to the Warfighter. The Offeror's milestone-driven IMS for capability delivery, culminating in an in-orbit delivery and performance per SDA's schedule and technical requirements, shall be integrated into this graphical representation.

4.2.2 CONOPS Overview

The Offeror shall provide a CONOPS for the operation of the T3 Tracking constellation, including network maintenance and initialization, spacecraft maintenance and on-orbit calibration, data transfer, and direct-to-Warfighter communications. The Offeror shall clearly identify any mission constraints or limitations to include radiation upsets and recovery plans.

4.2.3 Space Vehicle (SV) Overview

The Offeror shall present an overview of the SV, including but not limited to:

- SV block diagram
- Delta-V budget
- High-level mass budget
- Power budget with power modes
- Thermal management strategy
- Vehicle configuration showing all major components
- Backup TT&C system design
- Resiliency feature to protect against directed energy attack mechanisms

This section shall also include an overview of the SV concept of operations (CONOPS) including but not limited to a description of SV states and modes to be implemented in service of all required mission operations, and transitions between those states and modes.

A detailed Master Equipment List (MEL) with mass, power, Manufacturing Readiness Level (MRL), and Technology Readiness Level (TRL) or other flight heritage shall be included as an appendix and will not count against the Volume II page limit. The MEL is expected to include details of assemblies below the subsystem level. The power budget shall include the duty cycle and current best estimate for each component by power mode, as well as the orbit average power generated in each power mode and enough information to determine the battery depth of discharge.

4.2.3.1 Tranche 3 Tracking Layer Payload Architecture and Design

The Offeror shall present their overall payload architecture and design. For each payload element below, the Offeror shall include,but not be limited to a discussion of payload hardware, software, control, data transfer capability and/or limitations, and power requirements. Proposals shall demonstrate a clear understanding of each payload's mission application and the technical solution's ability to accomplish the tracking mission. Proposals shall also address any needed test and certification requirements and plans.

4.2.3.2 Infrared Mission Payloads

The Offeror shall provide a detailed description of the infrared mission payloads, including the projected size, weight, and power (SWAP) requirements, placement on the SV, maturity by component (TRL/MRL), technical capability, operational application, and image processing methods for the MW/MT infrared sensor and FC/MD infrared sensor (assuming differing designs). The Offeror shall describe the ground AI&T approach for each, including details of sensor calibration and environmental testing approach, and any radiation testing required. The Offeror shall provide a description of how on-orbit calibration, solar avoidance, continuous imaging operations, and pointing modes will be implemented. The Offeror shall propose how calibration will be completed on-orbit for initial operations and how it will be updated through the life cycle.

The Offeror shall propose one or multiple IR spectral bands based on SDA's stated mission objectives of detecting and tracking infrared targets (both bright and dim, and both static and moving), including hypersonic vehicles and other advanced missiles. The Offeror shall propose a detection and processing architecture for use on-board each Tracking SV and on the ground that:

a) performs sensor-level processing to calibrate data and update calibration solutions as needed,
b) suppresses backgrounds and detects candidate events to transmit to ground or Transport SV BMC3 units, and
c) generates 2D tracks to transmit to ground or Transport SV BMC3 units.

The Contractor shall also propose a fusion processing architecture that performs fusion of track data from multiple Tracking SVs from the vendor's own planes into 3D tracks. The 3D tracks will be formatted as standardized OPGA-8 messages to support interoperability within the MW/MT/MD community and Realtime Transfer Service (RTS). This function will support test, requirement verification, and algorithm tuning on the ground, both prior to launch and during operations. The Offeror shall describe their on-board data storage, lossless compression approach, and management of both raw and processed data for prioritization and routing based on the requirements of the ground IR MW/MT/MD mission.

The Offeror shall provide analysis supporting their design approach including, but not be limited to, the following:

- Trades analysis conducted to achieve optimal design performance
- IR mission data processing architecture
- IR Sensor configuration
- Spectral band choice
- Radiometry budget and accuracy
- IR sensor on-board data processing throughput and data management/storage approach
- Line-of-Sight budget and state vector accuracy
- Stray light management approach
- Target tracking performance (MDT, NET, coverage)
- Resiliency

4.2.3.3 Optical Communications Terminals (OCTs)

The Offeror shall provide a detailed description of the OCTs, including placement on the SV, individual and total fields of regard, and ability to create and maintain space-to-terrestrial and space-to-space links. The Offeror shall identify the anticipated power requirements, data rate, bit error rate, and any other parameters considered key by the Offeror. The Offeror should provide a link budget demonstrating the closure of the optical links and clearly state any assumptions used in the development of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB). The Offeror shall identify the compliance of their current system with the SDA OCT Standard v3.2, detail any non-compliant elements of their design, and describe the path to full compliance for those elements.

4.2.3.4 Ka Mission Payload

The Offeror shall provide a detailed description of K_a-band subsystems used for the mission payload communications. The Offeror shall identify the anticipated modulation and coding implementations, information rates, power requirements bit error rate, and any other parameters key to mission performance. The Offeror shall identify the antenna configurations, field of regard, field of view, ability to support tactical users, and deployment mechanisms for each of the antenna systems. The Offeror shall provide link budgets demonstrating the closure of required RF links and clearly state all assumptions used in the development of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB). The Offeror shall include a link closure analysis that substantiates the payload’s compliance with the requirement to concurrently support multiple tactical users over the required footprint under the specified link conditions.

4.2.3.5 Telemetry, Tracking, and Control (TT&C)

The Offeror shall provide a detailed description of the RF communications subsystems used for telemetry, tracking, and control (TT&C). The Offeror shall identify the anticipated modulation and coding implementation, information rates, power requirements, bit error rate, and any other parameters key to mission performance. The Offeror shall identify the antenna configurations, field of regard, and field of view supporting link closure at the required information rate, including the impact of blockages. The Offeror shall provide link budgets demonstrating the closure of required RF links and clearly state all assumptions used in the creation of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB).

4.2.3.6 Networking and Data Routing

The Offeror shall describe the networking capabilities enabled by their technical solution. The Offeror should detail the compliance of their system with the SDA NEBULA Standard v3.06 identify any non-compliant elements of their design and identify the path to full compliance for those elements. The Offeror should discuss their plans for on-orbit network initialization and provide estimates of system performance.

4.2.3.7 Red-Side Processing (RSP)

The Offeror shall describe the proposed approach to implementing a red-side processing capability to host dynamic network management applications, including which processing unit the capability is expected to be hosted on, the processing, memory, and storage resources reserved for hosted applications, and a description of how this capability meets all functional and interface requirements.

4.2.4 Spacecraft Bus

The Offeror shall present their SV design based on the manufacture or purchase of commoditized SV buses[1]. Should the Offeror elect to propose a non-commoditized bus alternative, they shall present their process for commoditizing the alternative bus in addition to detailed justification for their bus selection.

4.2.4.1 Bus Design and Heritage

The Offeror shall provide a detailed description of the bus including the size, weight, and power available. The Offeror shall present a discussion of the bus heritage to include:

- How many have flown and in what orbits
- Design life and experienced flight life
- Types and frequency of experienced anomalies and their solution(s)
- Any major modifications required to meet the T3 Tracking Layer objectives

If the proposed bus has not flown, the Offeror shall provide the heritage from an existing bus (if applicable) and the changes made to reach the proposed bus design along with the rationale for each proposed change.

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[1] A commodity spacecraft bus utilizes an existing design that can be easily manufactured and reused across multiple SVs and for multiple missions. A commoditized bus has been designed for manufacture and is suitable for, or currently being produced by, an assembly line-like process at scale.

For each of the major subsystems (e.g., electrical and power, attitude control, TT&C, etc.) the Offeror shall provide a detailed description of the subsystem as well as system heritage. Block diagrams are encouraged where they provide greater insight into the design approach. The Offeror shall discuss specialty engineering (e.g., radiation, electromagnetic interference, contamination control, etc.) issues and approaches as appropriate. The Offeror should discuss the overall fault management approach, including the safe mode and recovery implementation.

4.2.5 *S-band Backup TT&C System*

The Offeror shall provide a description of the backup TT&C system design and its expected performance, to include planned modulation and coding implementations, information rates, power requirements, bit error rate, and any other parameters key to mission performance. The Offeror shall identify the antenna configurations, field of regard, and field of view supporting link closure at the required information rate, including the impact of blockages. The Offeror shall provide link budgets demonstrating the closure of required RF links and clearly state all assumptions used in the development of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB).

4.2.6 *Directed Energy Resiliency System*

The Offeror shall implement some additional methods for enhancing resiliency of selected mission payloads against threats such as directed energy effects. The Government is interested in mature elements or components that enhance Tracking constellation resiliency, particularly the IR payload sensor. The Offeror shall provide a directed energy resiliency system that at minimum includes a directed energy threat warning sensor. The Offeror shall provide a feasible SV CONOPs that could be implemented to enhance survivability, thus providing the capability to continue IR mission operations during directed energy illumination. Additional information is available in the classified attachment (Attachment 3 – TRD, Appendix). The Offeror shall include SWAP, cost, integration, CONOPs, etc. for a resiliency approach.

4.2.7 *Cybersecurity*

The Offeror shall discuss their approach to implementing required Risk management Framework (RMF) controls and cybersecurity in their integrated space and ground system.

4.2.8 *Encryption*

The Offeror shall describe the proposed implementation of security concerns, such as red-black isolation, encryption, and command authentication. The Offeror shall describe the encryption systems and technologies used on the space vehicle and how these systems will obtain certification and interoperate with the systems of other PWSA vendors and the ground.

4.2.9 Manufacturability

The Offeror shall present their manufacturing and assembly processes, captured in digital format, to demonstrate their ability to produce high-quality repeatable fabrication of the SVs including:

- Assembly of key parts,
- Testing of individual parts and assemblies to ensure they meet the reliability and availability standards associated with the SV,
- Assembly and test of SV subsystems and payloads,

- Integration and test of subsystem hardware,
- Integration and test of software with the hardware subsystems. A key focus of this demonstration is the IR mission payload for assembly, integration, and test.

The Offeror shall describe their AI&T flow and how it supports the on-time delivery of the T3 Tracking SVs. The Offeror should explain their ability to procure or manufacture commoditized buses at low cost and high reliability while achieving economies of scale as demonstrated in the commercial sector. The Offeror shall further detail planned mitigations for any manufacturing-related risks cited in Volume I, including strategies for overcoming the inability of the Offeror or a Major Teammate to produce the required number of units. The Offeror shall provide estimates of the manufacturing capacity of their organization (if applicable) and any Major Teammate, as identified in Section 3, inclusive of current contracts and any potential contracts within the Tranche 3 timeframe.

If currently under agreement with SDA, the Offeror shall clearly and concisely document the ability to produce T3 Tracking SVs, to include all payloads, within manufacturing constraints imposed by ongoing SDA efforts.

4.2.9.1 Supply Chain

The Offeror shall address the production and supply chain capacity available for this program, including potential contingency plans. The Offeror will address supply chain security, and production scalability issues. SDA values vendor diversity to enable supply chain resiliency, please list expected vendors for key components including SV bus, dispenser, IR Payload, payload control electronics, image collection modules, IR Focal Plane Array, OCTS and cryptographic equipment.

4.2.10 Test, Demonstration, and Verification

The Offeror shall present their hardware and software, ground and orbit-based test and demonstration approach(es). This section should address all aspects of test and demonstration to include modeling, simulation, analysis, emulation, and ground and space-based testing. The Offeror should address requirements verification at the component, subsystem, system, vehicle, and constellation levels up to and through environmental tests. The Offeror shall specifically address requirements verification of the IR mission payload and image processing. The Offeror shall describe test approaches, methodologies, performance, verification, documentation, and staffing. This section should include sample test artifacts from previous programs, and identification of test platforms and hardware maintained by the Offeror. The Offeror shall address facility and hardware cleanliness requirements and contamination monitoring approach throughout AI&T and launch integration. The Offeror shall specifically address test approach for the IR mission payload to include calibration, image processing, and space environment survivability. The Offeror shall provide test approach for SV interface to ground segment.

For cases where test activities are expected to be accomplished outside of the Offeror's direct control, the Offeror should identify their approach to manage test conduct and the experience level of the organization the Offeror plans to use to conduct the test. Of particular interest to SDA is the Offeror's approach to conducting tests and demonstrations in concert with SDA prior to launch and during early orbit operations and checkout, as well as use of the SVs in conjunction with Warfighter exercises once in orbit. Throughout discussion of their test activities, the Offeror should clearly identify live, virtual, and constructive tests and demonstrations, if applicable, and any elements expected to be provided by the Government to facilitate completion of their verification and validation program.

4.2.11 Launch Services Overview

SDA will procure launch services using the NSSL program. The Offeror shall describe how nine (9) of their SVs will integrate onto a single dedicated launch vehicle, including any custom adapters or separation systems required to attach to the launch vehicle.

The Offeror shall describe the proposed integrated payload stack and launch configuration. This description shall include specification of any custom adapters or separation mechanisms required to attach the T3 Tracking payload stack to the co-manifested payload stack and in turn the launch vehicle. The Offeror shall specify the total mass of the SVs and any adapter systems and/or leave behind mass in each launch stack configuration. The Offeror should include graphics of each of the two payload stack configurations integrated onto representative launch vehicles. The Offeror shall clearly identify any limitations that their design imparts on the launch vehicle and specify any launch services (such as a T-0 dry nitrogen purge) or additional accesses required to support launch vehicle integration. If the proposed bus has flown on an NSSL class launch vehicle, the Offeror shall provide the bus heritage.

For pricing purposes, the Offeror shall assume that all T3 Tracking SVs-High launches will be conducted from Vandenberg Space Force Base, CA and all T3 Tracking SVs-Mid launches will be conducted from Cape Canaveral Space Force Station, FL.

4.2.12 Ground Segment Integration and Operations

The Offeror shall present their proposed approach to operating their constellation through their NOVA software. The Offeror shall present their approach to integrating their NOVA with the Government-provided SUPERNOVA. The Offeror shall define which functions their NOVA will provide, which functions are assumed to be provided by SUPERNOVA, and how the systems will interact during operations and maintenance activities. The Offeror shall describe their use of COTS software and tools and the resultant benefits to integrated operations. The Offeror shall describe how their ground system will command SVs and provide SV system health and status data to the integrated ground operations management systems for continuous situational awareness and performance monitoring. The Offeror shall describe the proposed design of their software-based NOVA system to include descriptions and functions of all external interfaces, including those that enable connectivity to the Offeror's factory. The Offeror shall identify the minimum requirements for the Government furnished cloud-based processing environment needed to host their proposed NOVA system to include the number of cores, amount of RAM, and amount of storage that must be available from a Government-provided hosting service or platform.

The Offeror shall describe their approach to developing and maintaining the SV Operational Capabilities and Limitations (SOCAL) model to ensure that SUPERNOVA has ready access to an accurate and current representation of the Offeror's SVs' performance and operational constraints. The Offeror shall describe how the SV NOVA will comply with its requirements for responsive mission plan feasibility assessments and responsive translation of approved mission plans to SV commands and subsequent upload of the commands for execution.

The Offeror shall describe the suggested approach for verification that the proposed SVs and NOVA are compatible with SUPERNOVA and the greater PWSA GS.

4.2.12.1 FlatSat(s) and SoftSat(s)

The Offeror shall describe the proposed design of their FlatSat(s) and SoftSat(s) and their planned use in support of system verification, test, and anomaly resolution. Mission Operations, Sustainment, and Disposal.

The FlatSat will reside at the Performer's facility and will include the hardware, software, and firmware for the satellites avionics modules and payloads as if it were in connected in an actual space vehicle. The FlatSat will support integration and testing and anomaly resolution.

Sensor Model Plug-ins (SMPs) will be provided by the T3 SV Performers and will emulate the SV and IR mission payload (e.g., orbit position, timing, data outputs) and necessary spacecraft functions to enable SUPERNOVA to NOVA end-to-end testing on the ground. The intent of the SMP is to provide a software emulation of OPGA-79 products and other related mission data output products in sufficient detail to support the multi-sensor analysis on the ground. The SMP is not responsible for simulating the receipt of SV commands and generating simulated SV telemetry. The intent of the telemetry output of the SMP is to mimic the total mission data output rate from each SV. The SMP is expected to emulate SV Position, Nav, satellite cross links and/or GEP connections as needed to produce OPGA-79 messages that represent expected on orbit performance of the IR payload. The SMP implementation can use any combination of OBP algorithms or statistical models to generate OPGA-79s, so long as the resulting OPGA-79s meeting the functional and performance characteristics of the T3 sensor(s) to be developed.

A SoftSat microservices framework will be provided to the T3 SV performers GFE and developed by the Ground Segment integrator in collaboration with T3 SV performers. Many instances of SoftSats (framework + SMP) will execute in a cloud environment to emulate the T3 tracking constellation in support of system integration and test activities to be conducted by the Ground Segment integrator.

The Offeror shall identify the support that they will provide to T3 Tracking operations during nominal constellation operations, surge support, and anomaly operations. The Offeror shall describe their operations concept, level of automation, and staffing requirements for nominal operations. The Offeror shall describe their approach for anomaly resolution, deficiency reporting, and planned and unplanned system maintenance. The Offeror should list the documentation and training anticipated to be provided to SDA operations personnel. The Offeror shall describe the disposal plan for each SV and how that plan will reliably dispose of decommissioned SVs.

4.2.13 Mission Operations, Sustainment, and Disposal

For MW/MT submissions, the Offeror shall provide a CONOPs for the operation of the T3 Tracking Layer Constellation for persistent surveillance, warning, detection, and tracking of static and moving infrared targets including conventional and advanced missile threats such as hypersonic missile systems. The Offeror shall include CONOPs for the collection, processing, and transmission of MW/MT system Tracking data. The Offeror should include CONOPs for other applicable missions appropriate for their design. The Offeror should clearly identify any mission constraints or limitations. The Offeror shall identify the support that they will provide to T3TRK operations during nominal constellation operations, surge support, and anomaly operations. The Offeror shall describe their operations concept, level of automation, and staffing requirements for nominal operations. The Offeror shall describe their approach for anomaly resolution, deficiency reporting, and planned and unplanned system maintenance. The Offeror should list the documentation and training anticipated to be provided to SDA operations personnel. The Offeror shall describe the disposal plan for each SV and how that plan will reliably dispose of decommissioned SVs.

For MWTD submissions, the Offeror shall provide CONOPs for implementing their missile defense capability and shall address the CONOPs for processing and transmission of quality missile defense tracks. Offerors may assume the target's trajectory remains within the field of regard of a single MWTD SV. The Offeror shall identify the support that they will provide to T3TRK operations during nominal constellation operations, surge support, and anomaly operations. The Offeror shall describe their operations concept, level of automation, and staffing requirements for nominal operations. The Offeror shall describe their approach for anomaly resolution, deficiency reporting, and planned and unplanned system maintenance. The Offeror should list the documentation and training anticipated to be provided to SDA operations personnel. The Offeror shall describe the disposal plan for each SV and how that plan will reliably dispose of decommissioned SVs.

4.2.14 Appendices

The following appendices shall be included as attachments to Volume II, subject to the page limitations as indicated, and will not be included in the Volume II page limit. Any pages beyond the specified limit will be removed from the volume and will not be evaluated.

4.2.14.1 Radiometry Analysis

A detailed radiometry analysis shall be provided as an Appendix for the IR payload. The Offeror shall demonstrate the Minimum Detectable Target (MDT) and Noise Equivalent Target (NET) performance of the T3 Tracking system, as well as probability of detection, tracking, and warning, as specified in the classified TRD SOW appendix. The Offeror shall detail the signal, noise, and clutter parameters and performance expectations in a radiometry budget or equivalent to allow the Government to evaluate the risk associated with the sensor and processing chain performance.

4.2.14.2 Metric Analysis

The Offeror shall demonstrate coverage and state vector accuracy performance expectations of the T3 Tracking system and provide a line-of-sight knowledge budget, including random and bias terms and contributing parameters and assumptions, to allow the Government to assess the state vector accuracy performance.

4.2.14.3 Mission Availability and Reliability

The T3 Tracking Technical Requirements Document includes notional requirements for the availability and Mean Time to Recovery (MTTR) of the mission functionality. The Offeror shall provide a detailed, quantitative assessment of the performance of their proposed design and the features that enable that performance. If the Offeror wishes to propose other availability and reliability requirements, those rationale for those requirements shall be supported with analysis demonstrating the mission impacts that result from the proposed modifications. The Offeror shall also propose an approach for verifying the availability and reliability requirements prior to launch, as well as maintaining and updating availability performance estimates on orbit.

This appendix shall not exceed five (5) pages.

4.2.14.4 Commercial Data Sheets

The Offeror shall submit commercial data sheets for any major subsystem identified as COTS, as defined in Section 3. Additionally, the Offeror may submit up to ten (10) commercial data sheets for other systems, subsystems, or components. Commercial data sheets shall be submitted as a list of hyperlinks to publicly available, downloadable .pdf data sheets on the Offeror's websites or their subcontractors' websites. Each downloadable .pdf file shall be no more than two (2) pages. Links will be accessed in the order provided in the proposal and only the first twenty (20) pages beyond those associated with major subsystems cited as COTS will be considered. Any pages beyond the twenty (20)-page limit will be discarded.

4.2.14.5 Parameter Specification

The Offeror shall complete the value column in each worksheet of the Tracking Layer KeyParameterSpecification.xlsx file in the Bidder's Library and shall include the completed .xlsx file as an appendix to Volume II. This file shall be submitted as a Microsoft Excel (.xlsx) file. Parameter specifications submitted as .pdf files will be discarded. All values in the table shall be provided in the units specified. Upon selection for award and at the discretion of the Government, the proposed values in this table may be included in the final SOW and TRD in lieu of the current threshold and objective values specified in Attachment 1 – SOW and Attachment 3 – TRD.

4.2.14.6 Technical Requirements Verification Methods

The Offeror shall submit a version of Attachment 3 – TRD as an appendix to Volume II, with a column appended to specify the Offeror's proposed verification method for each requirement. If the proposed verification method differs from the Government's recommended verification method, the Offeror shall include rationale for the proposed alternative method in an adjacent column.

4.3 Volume III – Integrated Master Schedule

Volume III shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable **content, other than those specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated.**

4.3.1 Integrated Master Schedule (IMS)

The Offeror shall present the milestone-driven IMS designed to meet the Government's notional milestones provided in Section 2.1 of Attachment 1 – SOW and launch schedule identified in Section 4 of Attachment 1 – SOW. The Offeror shall clearly indicate the anticipated schedule margin for each launch, where margin is calculated as the number of calendar days between the date of completion of the final SV for the launch and the corresponding delivery date to the launch site, as specified in Attachment 1 – SOW. The Offeror shall clearly indicate allocated margin against key milestones to indicate additional schedule margin planned. The Offeror shall identify the dates for all major reviews and program milestones and include entrance criteria, key accomplishments, and exit criteria for each milestone. If the milestones proposed correspond to payments anticipated by the Offeror, this Volume should so state and link to the appropriate section in *Volume IV – Price and Rationale* so that the milestone payment appears in *Volume III –*

Integrated Master Schedule and the payment amount, terms, conditions, and Government acceptance criteria for payment appear in *Volume IV – Price and Rationale*.

At least two (2) of the pages shall be a separable, standalone detailed Gantt chart (or similar illustration) clearly presenting the Offeror's milestone-driven integrated master schedule, including key program milestones and critical paths. This chart should include dependencies, assumed Government or Major Teammate hardware, software, or information deliveries and/or approvals, tests, demonstrations, reviews, and/or any other key activities in the Offeror's proposed program.

The Offeror shall clearly delineate the primary and secondary critical paths within their schedule and any additional critical paths deemed appropriate to justify their ability to deliver the proposed SVs on time and on cost. The Offeror shall discuss the critical path analysis used to determine the primary and secondary critical paths. The Offeror shall discuss their approach to managing program progress against the baseline schedule. The Offeror shall include delivery dates of components from all Major Teammates and shall clearly identify which of these components lie on the critical path.

The Offeror shall identify maturation and risk reduction plans for any SV components that are at a TRL of less than six (6) at the time of the proposal. Key technical events in the maturation of these technologies should be clearly presented in the IMS appendix. The Offeror shall elaborate upon their approach(es) to mitigate schedule risks identified in Volume I and ensure on time delivery. The Offeror shall identify which teammate deliveries and which external dependencies produce the most risk to on time delivery.

The Offeror shall deliver their complete IMS in Microsoft Project-compliant native format as well as .pdf format in conjunction with this volume. The complete IMS shall include schedule details down to at least Level 3 of the work breakdown structure (WBS) and details to at least the subassembly level for high-risk areas and to the subsystem level elsewhere. IMS content shall be tailored to address each WBS element's unique design, development, assembly, integration, and test activities, as applicable.

The delivered IMS files (Microsoft Project and .pdf format) will not count against the Volume III page limit, however, the Gantt chart is included in the page limit.

4.3.2 Schedule Risk

The Offeror shall clearly and concisely identify schedule risks based on their approach, design, team construct, and/or consolidated backlog. For each risk, the Offeror shall identify key mitigation steps as well as internal and external dependencies of those steps.

4.3.3 Appendix – Portfolio IMS

If the Offeror (Prime or Major Sub) is currently providing SVs, ground systems or other materiel to SDA, they shall submit a Portfolio IMS as an Appendix to Volume III. This Portfolio IMS shall demonstrate how multiple programs will be managed.

At least two (2) pages shall be a separable, standalone detailed Gantt chart (or similar illustration) clearly presenting the Offeror's milestone-driven portfolio-level integrated master schedule, including key program milestones and critical paths. This chart should include dependencies, assumed Government or Major Teammate hardware, software, or information deliveries and/or approvals, tests, demonstrations, reviews, and/or any other key activities in the Offeror's proposed SDA portfolio. The Offeror shall identify portfolio-level schedule risks, their mitigations, and succinctly discuss dependencies of one program upon another and/or how each program will be completed successfully even if another program in the portfolio encounters schedule, technical or programmatic difficulties.

4.4 Volume IV – Price and Rationale

Volume IV has no page limit.

4.4.1 Executive Summary and Pricing Rationale

The Offeror shall detail their pricing assumptions and rationale that guided their firm fixed price (FFP) submission. The Government needs to understand the Offeror's assumptions in building their final price such that any discussions deemed of value can be conducted. The Offer should discuss any constraints on the Government such as time to review and/or approve contractual documentation, constraints on mission partners such as time to receive needed data or information, etc. Where appropriate, the Offeror should discuss their use of learning in driving down subsequent SV costs when producing more than a single unit and where such learning has led to cost savings on previous programs.

4.4.2 Price Proposal Structure

The Offeror's price proposal shall comply with the guidance in the following sections. The Offeror may assume an Authorization to Proceed (ATP) of 15 October 2025.

4.4.2.1 Bidding Unit

The Offeror shall provide the price for one (1) full unit, where a unit consists of:

- Two (2) orbital planes of eighteen (18) T3 Tracking Layer SVs and system-level program management and systems engineering
    - One (1) plane at ~45-degree inclination and ~1020 km altitude and one (1) plane at ~81-degree inclination and ~1000 km altitude
- SV to LV integration for two (2) launches (launch stack integrates nine (9) SVs
- All Approved Development Units (ADU) for interoperability testing
- Required NOVA instances and integration into GS
- FlatSat
- Sensor Model Plug-ins and supporting documentation required for Operations SoftSat
- Two (2) years of operations and sustainment following the second plane's Functional Acceptance Review (FAR) as part of base contract
- Three (3) one (1) year options of operations and sustainment following the base period of this agreement
- All other efforts necessary to meet the requirements as specified in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document

4.4.2.2 CLIN Structure

The Government anticipates awarding to a single CLIN. The CLIN will be awarded as Firm Fixed Price (FFP). All price details of the CLIN shall be included for a compliant proposal. The Offeror shall propose milestone payment structures in accordance with the instructions in the following sections. This information will be used by the Government to evaluate the reasonableness of, and risks associated with, the proposed technical solution, schedule, and price.

4.4.2.3 Work Breakdown Structure (WBS)

The Offeror shall provide a WBS, that is compliant to the Level 3 WBS shown in Figure 4-1. The Offeror may include work breakdown details below the levels indicated in Figure 4-1 at their discretion.

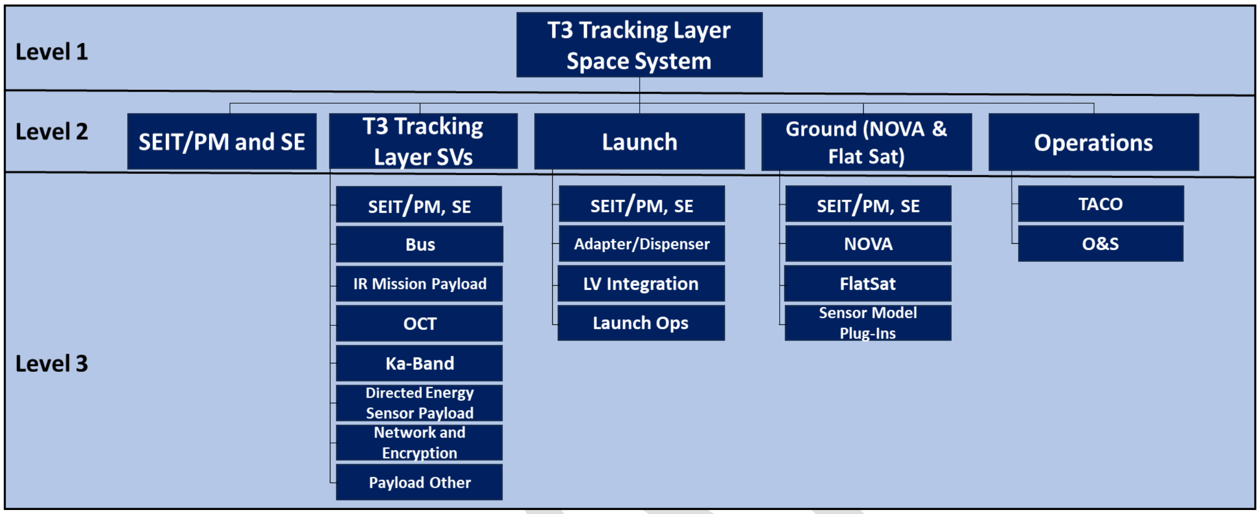

Figure 4-1 Level 3 WBS

4.4.2.4 Milestone Payments

The Government intends to fund this FFP effort via event-based Milestone Payments. The Offeror shall propose a Milestone Payment Plan compliant to guidance in this section.

The Offeror shall propose a milestone payment structure by clearly identified program events with supporting rationale for their selection. In addition to using the format specified in Attachment 2, the Offeror shall also provide their milestone payment structure in Microsoft Excel format (.xslx). For each milestone payment, the Offeror shall specify the payment amount, terms, conditions, Government acceptance criteria, and percentage of the total price for 18 SVs. The Offeror's payment plan shall adhere to mandatory milestones and minimum payments listed in Table 4-2 but may otherwise add, delete, or modify in areas where not specified.

Table 4-2. Sample Milestone Payments

Milestone	Mandatory	Minimum % of Total Price
Kickoff	No	---
SRR/SDR	Yes	5%
PDR	Yes	5%
CDR/PRR	Yes	9%
Test Readiness Review (TRR)-Calibration	Yes	5%
System TRR	Yes	5%

Optical Communications Terminal Interoperability Test (OIT) Success	Yes	4%
S-Band TT&C and Ka-Band Communications Systems Compatibility Test (SKCT) Success	Yes	4%
Networking and Encryption Interoperability Test (NEIT) Success	Yes	4%
Pre-Ship Review (PSR) - First Plane	Yes	1%
PSR - Second Plane	Yes	1%
Functional Acceptance Test Readiness Review (FATRR) – First Plane	Yes	4%
FATRR – Second Plane	Yes	4%
Functional Acceptance Review (FAR) - First Plane & IR calibration complete	Yes	5%
FAR - Second Plane & IR calibration complete	Yes	5%

Additional milestones that do not have associated minimum percentage of Total Price specified can be found in Attachment 2 – Schedule of Milestones and Payments.

The Offeror's payment plan and expected obligations should adhere to the payment plan according to Table 3-4 (Tracking SVs and SEIT/PM) and should not exceed the cumulative values provided in Table 4-3 by Government Fiscal Year.

Table 4-3 Payment Schedule

Government Fiscal Year	Cumulative % of Total Price
FY26	20%
FY27	50%
FY28	80%
FY29	100%

The Offeror's payment plan should also avoid milestone payments scheduled to occur between 1 October and 15 November of every calendar year.

4.4.2.4.1 Delivery Incentive

In addition to the negotiated price and milestone payment plan, the Government offers a delivery incentive for the successful, on-time delivery of the entire bidding unit. This incentive fee is only offered in full based on successful completion of the following terms and conditions:

- Delivery of the first plane of 9 SVs, integrated to the LV adapter, to the designated launch processing facility no later than 2 Mar 2029
- Delivery of the second plane of 9 SVs, integrated to the LV adapter, to the designated launch processing facility no later than 1 Jun 2029
- All SVs, and associated ground systems, are fully compliant with the system technical requirements as specified in Attachment 3 – Technical Requirements Document
- Successful completion of Functional Acceptance Review (FAR) for all delivered SVs and ground systems

Failure to meet any of the terms and conditions of the delivery incentive may result in forfeiture of the entire value, however, the Performer is still eligible to complete agreement performance, collecting all milestones payments, as appropriate and satisfied. The delivery incentive is fifteen (15) million dollars ($15,000,000.00), payable upon successful completion of all terms and conditions.

The Government will not accept differing or reduced proposed terms or conditions as part of this solicitation, during agreement negotiations with successful Offerors, or during agreement performance with awardees.

4.4.2.5 On Orbit Performance Penalty Payment

The Offeror shall propose a maximum liability of the agreement value at risk for potential on-orbit performance using the Table 4-4 below. The Maximum Liability Value for each period shall be 1/10th of the Maximum Liability proposed and shall be included in the Attachment 2 – Schedule of Milestone Payments, On Orbit Performance Section.

Table 4-4 – On Orbit Performance Penalty Payment

On Orbit Performance Period	Estimated Assessment Date	Maximum Liability $ Value	Performance Assessment Score Percentage	Performance Penalty Payback Value
1	TBD			
2	TBD			
3	TBD			
4	TBD			
5	TBD			
6	TBD			
7	TBD			
8	TBD			
9	TBD			
10	TBD			
Total Potential Performance Liability				

4.4.3 Proposal Pricing Details

The Offeror shall provide the total price and elements of price broken down by WBS, Government Fiscal Year, Major Teammates, and milestone payment schedule, as follows:

1. Price by work breakdown structure (WBS) task and Government Fiscal Year (GFY) using the template in Table 4-5. The price shall be provided for items down to Level 3 of the WBS for each GFY.

Table 4-5. Template for Pricing of Tasks by WBS and Government Fiscal Year

WBS #	WBS Element	GFY26	GFY27	GFY28	GFY29	GFY30	GFY31	GFY32	GFY33	Total

2. A summary of projected funding requirements by month and year per Government Fiscal Year to enable fiscal and milestone payment planning. Funding requirements for a given fiscal year shall equal the milestone payment requirements in that fiscal year.
3. Price and percentage of price by the prime and Major Teammates, where the minimum set of Major Teammates is defined in Section 3.
4. Milestone payment schedule based on the template provided as Attachment 2 – Schedule of Milestones and Payments.

Each price breakdown shall also be provided in Microsoft Excel (.xslx) format. The Government expects the total prices by GFY and grand total price as presented in each of the above breakdowns to be the same values in each breakdown.

The Offeror shall identify pricing assumptions of which may require incorporation into the resulting award instrument (e.g., use of Government Furnished Property/Facilities/Information, access to Government Subject Matter Experts, etc.) for all members of the team.

For the purposes of pricing, the Offeror shall assume that the Test and Checkout Center is located near one of the SDA Operations Centers, which are located on Grand Forks AFB, ND and Redstone Arsenal, Huntsville, AL.

4.4.3.1 Additional Pricing Data

The value specified as the Total Price shall represent the total price of the proposal for one bidding unit and will be evaluated by the Government for price reasonableness during source selection.

The Offeror shall provide a summary of Pricing in accordance with Table 4-6.

The base proposal shall propose the cost of two (2) planes of nine (9) SVs as one bidding unit, along with the launch, Ground Segment, and separately identify the delivery incentive. The Government reserves the right to award a third plane.

Table 4-6 Pricing Table

Item	Description	Price
1	18 T3 Tracking SVs	Price of first plane of 9 T3 Tracking Layer SVs
		Price of second plane of 9 T3 Tracking Layer SVs
		Price of system level SEIT/PM and SE for 2 planes of T3 Tracking SVs

2	SV to LV Integration – Launch 1	Price of SV to LV integration for the first plane of 9 T3 Tracking Layer SVs
3	SV to LV Integration – Launch 2	Price of SV to LV integration for the second plane of 9 T3 Tracking Layer SVs
4	Contractor Ground Elements (NOVA)	Price of ground hardware and software for SV and constellation management for SVs
5	FlatSat	Price of FlatSat hardware and software for SV constellation management for SVs
6	Sensor Model Plug-ins (SMP)	Price of SMP and supporting documentation for Operations SoftSat for SVs
7	Operations and Sustainment (O&S) 2-Year Base Period	Price of O&S for 2 planes of SVs for 2-year base period, includes LEOPs
8	O&S Option Year 1	Price of O&S for 2 planes for first option year
9	O&S Option Year 2	Price of O&S for 2 planes for second option year
10	O&S Option Year 3	Price of O&S for 2 planes for third option year
11	Delivery Incentive	$15,000,000.00
12	Total Baseline Price	Total Price: Sum of proposed summary prices

4.4.4 Government Furnished Equipment, Information, and Property

The Offeror shall compile a consolidated list of ALL Government furnished equipment, information or property being requested by themselves and **all** team members for ease of Government evaluation. This list shall clearly delineate the item, need date and the risk associated with non or late delivery on overall program schedule, cost, or technical performance. The Offeror may use the list provided by the Government in Attachment 5 – Government Furnished Items (GFX) List as the starting point for their proposed list of Government-furnished content.

4.4.5 Major Teammate Proposals

Offerors are responsible for compiling and providing all Major Teammate proposals for the primary agreements officer (PAO), where a Major Teammate is defined in the introduction to this volume. Teammate proposals should include all Intercompany Work Transfer Agreements (IWTAs) or similar arrangements. All proprietary teammate proposal documentation shall be prepared at the same level of detail as that required of the Prime. Any such documentation that cannot be provided with the proposed Prime performer's proposal shall be provided to the Government either by the Prime performer or by the teammate organization via DoD SAFE, following the instructions in Section 3.2.1

Teammate proposals will only be used to determine the price reasonableness of Offerors' proposals. No information included in any teammate proposal will be evaluated for the purpose **of satisfying the requirements as stated in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document or meeting the instructions associated with Volumes I, II, III, and V of this solicitation.**

4.5 Volume V – Experience and Past Performance

The page limit for Volume V depends on the number of past performance citations as stated below. Any pages in excess of this limit, other than those noted below as specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated. Classified past performance citations may be submitted as attachments to Volume V within the specified page limits.

4.5.1 Executive Summary

The Offeror should introduce the current and past performance citations, clearly stating why they have chosen the efforts and how they best prove their ability to perform the proposed work scope. Each citation should provide specific, measurable validation of the Offeror's rationale and document successful current and past performance directly relevant to the current solicitation. The Government expects direct correlation between the current and past performance citations and the résumés submitted in *Volume I - Governance*. Include references to résumés in Volume I for any classified personnel experience specifically to identify key personnel and their role in the classified effort.

4.5.2 Recent and Relevant Past Performance

The Offeror should submit at least one (1) and may submit up to four (4) separate efforts by the Prime and at least one (1) and up to two (2) separate efforts for each Major Teammate. These submissions should clearly demonstrate the ability of the Offeror and its teammates to satisfy the totality of the Tranche 3 Tracking Layer requirements. The minimum set of Major Teammates is defined in Section 3. All submissions shall have a period of performance within five (5) years preceding the proposal due date. Significantly more relevance will be assigned to efforts with technical accomplishment during the five (5) year timeframe as opposed to only operations or support activities occurring within that timeframe.

For each submitted citation, the Offeror should identify:

- Customer (Name, phone, email, affiliation)
- Scope to include number of SVs/payloads/systems/subsystems/components previously delivered or to be delivered
- Place of performance, period of performance, agreement type
- Bid cost/price, currently invoiced cost/price, final price/estimate at completion
- Specific rationale for effort selection and detailed justification of relevance
- Schedule, technical and/or programmatic issues encountered, mitigations identified and undertaken, and positive or negative results seen as a result of mitigation efforts

The Offeror is highly encouraged to use key graphics to convey important elements of their current and past performance. High level graphical descriptions of the operational concept (e.g., DoDAF's OV-1) depicting numbers and locations of SVs, Gantt charts showing key program milestones planned and achieved, Earned Value charts (if applicable) showing estimates and current status, etc. are all *substantially* more valuable to the Government for relevance evaluation purposes than detailed written explanations. These artifacts should be taken directly from the efforts cited and be suitable for use by the Government in conjunction with interviews which may be conducted with the customers named in each citation.

4.5.1.1 Citations

Project citations shall not exceed four (4) pages each and should stand alone by project.

For each Past Performance citation, the Offeror shall provide the Attached Past Performance Questionnaire (Attachment XX) to the Government or industry contracting officer or program manager to complete and submit directly to the SDA in accordance with the questionnaire instructions. The Offeror shall provide a list of names and point of contact for the PPQs in the proposal. (Note PPQs completed by the Government and the PPQ POC list are not included in the page count).

Offerors are hereby notified that SDA intends to survey Contractor Performance Assessment Reports (CPARs) and to conduct interviews with customers identified in *Volume V*. Information obtained via either of these methods will be included in Offeror evaluations per the criteria in Section 4.4.

4.6 Volume VI – Draft Other Transaction (OT) Agreement

The Offeror shall provide a proposed OT agreement, including all of its attachments, using the attached draft OT template included with this solicitation.in MS Word format. Any proposed changes to the draft OT template by the Offeror shall be indicated with the Track Changes function and rationale provided for the proposed change.

5 Submission Instructions

SDA requires that Offerors submit only electronic proposals. Unless otherwise stated in these instructions, all Volumes along with any appendices and/or attachments shall be submitted only in .pdf format and must be submitted independently by file as SDA will not accept .zip files. Offerors shall submit their responses via DoD Secure Access File Exchange (SAFE): https://safe.apps.mil. Offerors shall submit one (1) original copy of each Volume and appendices and/or attachments NO LATER THAN 1700 ET 21 May 2025. Offerors shall submit each their proposal via DoD SAFE submission and all files should be included in a single DoD SAFE submission. Each file submitted must be clearly labeled with the SDA solicitation number, Prime performer organization name, Prime performer proposal title (short title recommended), and Volume number and appendix or attachment number (as appropriate). The size limit for each SAFE file transmission is 8192 MB.

Offerors shall request a SAFE Drop-off Request, via an email to osd.pentagon.ousd-r-e.mbx.sda-ps-23-05@mail.mil NO LATER THAN 1200 ET 14 May 2025. The Offeror shall request a SAFE Drop-off Request regardless of whether or not they have access to DoD SAFE already. The subject line shall be: T3 Tracking Layer Solicitation Submission SAFE Request (m of n) and the Offeror's organization name, where "m" is the request number and "n" is the total number of requests that will be made by the Offeror. If the Offeror's submission will exceed the DoD size limit, then they shall make the required number of requests to complete their submission. Within the body of the email, the Offeror shall clearly state the email address to which the Drop-off Request will be sent. The Offeror is free to request read receipts for each Drop-off request email sent. The Offeror will receive an email with their SAFE Drop-Off Request NLT 1200 ET 15 May 2025. Note that this email will not come from the email address listed above.

Within the DoD SAFE submission form, the Offeror shall check both the "Encrypt every file" check box and the "Send me an email when each recipient picks up the files" check box. The password shall be sent under separate email to the email address above with the subject line: T3

Tracking Layer Solicitation Submission SAFE Request (m of n) Password and the Offeror's organization name. The "Short note to the Recipients" field shall clearly identify the Offeror's organization name, the proposal name, and the number of the submission.

Proposals received after 1700 ET 21 May 2025 will be considered late and not evaluated. Offerors whose proposals are considered late will be so informed by the SDA Agreements Officer and their proposals will be deleted without any files being opened or read. Classified appendices and/or attachments may be submitted if required. **Should a classified submission be required, the Offeror must contact SDA via the email address above for submission instructions.**

5.1 Submission of Major Teammate Proposals

Major Teammate proposal documentation that cannot be submitted directly by the Offeror shall be submitted by the teammate via DoD SAFE in accordance with the instructions in Section 4.3. In addition to the details specified in that section, the email request for a SAFE Drop-off request shall include the teammate organization name in the subject line of each email. Each file submitted must be clearly labeled with the SDA solicitation number, Prime performer organization name, Prime performer proposal title (short title recommended), teammate organization name, and Volume number and appendix or attachment number (as appropriate). Teammates submitting separate documentation via DoD SAFE must send an email to the address in the Submission Instructions notifying SDA of this submission method **NO LATER THAN 1200 ET 20 May 2025.**

6 Proposal Evaluations

The Government will evaluate offers using a competitive best value trade-off process among price and the non-price Factors listed herein. Thus, the Government may elect to award to other than the lowest priced Offerors, or other than the highest technically rated Offerors. The Offeror's initial proposal should contain the Offeror's best terms regarding price and technical capability.

In making its award decision, the Government will use the evaluation Factors described in the following sections. The Offeror should not assume the Government will give credit for any capability or knowledge unless it is specified in the proposal. The Government may reject any or all proposals, if such action is in the best interest of the Government and waive informalities and minor irregularities in proposals received.

While the Government strives for maximum objectivity in its evaluations, the source selection process, by its nature, is subjective, and, therefore, professional judgment is implicit throughout the entire process.

During the best value tradeoff process, the Government may also consider technical and programmatic risk across its portfolio of programs and contracts, both SV Prime performers and key mission payload providers, when making the selection determination. Additionally, the Government values diversity among Prime performers and key mission payloads.

6.1 Evaluation Methodology

The following Factors will be used to evaluate each proposal. The basis for evaluation for all Factors will be the Offeror's full proposal. Factors 1–5 are listed in descending order of importance. When combined, Factors 1-5 are approximately equal to Factor 6, Price.

The Government will evaluate information that the Offeror(s) provide in response to Section 4 Proposal Preparation Instructions. Proposals must conform to this solicitation. Any submitted proposal that does not adhere to the requirements in this solicitation may be eliminated from the competition and become ineligible for award.

Offeror(s) are required to meet or exceed all solicitation requirements and technical requirements or to explicitly identify anticipated waivers, in addition to those identified as Factors. Failure to comply with the terms and conditions of the solicitation ***may*** result in the Offeror(s) being ineligible for award. The Government reserves the right to cross-reference between volumes and sections of volumes to support proposal evaluation.

6.1.1 Factor 1 – Schedule

For Factor 1 – Schedule, the Government will assess the Offeror's proposed schedule and their demonstrated ability to satisfy all requirements contained in the Attachment 1 – SOW and Attachment 3 – TRD, in a timely fashion.

The basis of the assessment will be the Offeror's schedule and schedule margin as detailed in Volume III – IMS of their proposal, the Master Equipment List (MEL), the Manufacturability section of proposal Volume II – Technical. The Government may also consider any schedule-related information provided within any of the other proposal volumes. Identified strengths and weaknesses in technical or management areas that affect schedule reasonableness or realism or contribute to schedule risk will also be considered in this assessment.

The Government will evaluate the schedule reasonableness and realism based on

- Offeror's demonstrated understanding of tasks and activities required to accomplish the work
- The estimated duration of tasks and activities based on technical and manufacturing/production maturity
- The sequencing of these tasks and activities
- The identification of any dependencies and other schedule relationships (e.g., subcontractor/supplier delivery dates and delivery dates of Government- furnished material)
- The critical path identification and analysis,
- The identified schedule risks and risk response strategies
- The proposed milestones, and performer and/or Government decision gates
- The schedule performance criteria used to manage progress against the integrated master baseline schedule.
- Evaluated strengths and weaknesses in technical or management areas that affect schedule reasonableness or realism or contribute to schedule risk

For those Offerors currently providing hardware, software or other solutions to SDA based on existing efforts, key elements of the overall schedule risk evaluation will be based on a portfolio level assessment of existing and proposed work.

6.1.2 Factor 2 – Technical Solution

For Factor 2 – Technical Solution, the Government will assess the Offeror's demonstration that their technical solution will meet the system requirements in Attachment 1 – SOW and Attachment

3 – TRD. Evaluation credit may be given for solutions exceeding mandatory thresholds. The technical solution includes but is not limited to, the following:

A. The satellite bus
B. IR Mission Payloads and required interfaces
C. IR Mission space and ground data processing architecture and performance
D. Other mission payloads and their required interfaces
E. Telemetry, Tracking, and Command capabilities
F. Satellite system and subsystem integration and testing
G. External system interoperability
H. All Offeror-provided ground elements and their interfaces to external ground systems
I. Multi-vehicle integrated stack and launch vehicle integration
J. Launch and Early Operations (LEOPs) and,
K. Operations and sustainment.

The basis of the assessment will be the Offeror's proposed technical solution and its mission utility, as provided in proposal Volume II – Technical and its appendices and attachments.

The Government will evaluate the technical solution based on the fidelity of technical details provided on the technical design, evidence-based Technical Readiness Level (TRL) and Manufacturing Readiness Level (MRL) assessments for subsystem or components, analysis or engineering products demonstrating the proposed design can meet Attachment 1 – SOW and Attachment 3 – TRD specifications, and results of any risk reduction efforts to date or detailed plans for risk mitigation during program execution.

6.1.3 Factor 3 – Past Performance

For Factor 3 – Past Performance, the Government will assess the Offeror's probability of meeting the solicitation requirements based on the Offeror's demonstrated record of work performance.

The Government will evaluate Factor 3 based on information provided within proposal Volume V – Past Performance. The Government may consider past performance information not provided by the Offeror(s) including, but not limited to:

- Contractor Performance Assessment Reporting System (CPARS)
- Publicly available information
- Federal Awardee Performance and Integrity Information System (FAPIIS)
- Electronic Subcontract Reporting System (eSRS)
- Interviews with Customer Program Managers, Contracting Officers, Fee Determining Officials, or other personnel with managerial or oversight responsibilities related to the cited effort
- Past performance questionnaires completed by customers from whom the Offeror has received previous awards

The Government will evaluate the Offeror's recent (performance was within five (5) calendar years of the proposal due date) and relevant record of past performance information obtained to determine whether past performance efforts relate to the scope of activities described in the Attachment 1 – SOW and Attachment 3 – TRD.

The Government will review the past performance information collected and determine the quality of the Offeror's performance, general trends, and the usefulness of the information and incorporate these into the performance confidence assessment. **More relevant past performance will have more influence on the performance confidence assessment than past performance of lesser relevance**. In evaluating Past Performance, the Government will consider the Offeror's previously demonstrated:

- Past performance, history, and experience.
- Ability to accomplish requirements to receive milestone or performance-based payments. This includes activities such as interim deliverables, delivery of CDRLs or other technical and business reports, and completion of customer direction through task assignments or technical directions.
- Ability to meet technical requirements and performance standards for previous work.
- Ability to meet delivery or performance dates.
- Approach in determining probable root cause for less than fully successful missions and resultant actions taken to improve reliability.
- Safety and mission assurance performance record, including any safety mishaps and associated resolution.

The Government will consider recent or on-going SV development contracts or agreements with SDA as very relevant and, therefore, Performer performance on those efforts will be heavily considered and weighted in the evaluation of Factor 3 even if not specifically submitted by the Offeror or any Major Teammate.

6.1.4 Factor 4 – Technical Management and Processes

For Factor 4 – Technical Management and Processes, the Government will assess the Offeror's overall management capability and demonstrated ability to meet the contract work performance requirements in Attachment 1 - SOW and Attachment 3 – TRD.

The basis of the assessment will be the Offeror's proposed technical management and processes approach as provided in proposal *Volume I – Governance* and its appendices and attachments, the Manufacturability section of proposal *Volume II – Technical*. The Government may also consider any management-related information provided within any of the other proposal volumes.

The Government will evaluate the Technical Management approach based on:

- Demonstrated efficient and effective management methodologies, processes, and procedures to achieve SOW and TRD objectives, including flow-down to subcontractors
- Demonstrated capabilities of key personnel who will enact the plan
- Demonstrated ability to provide Government insight into progress, risks, and issues
- Demonstrated the ability to execute the SOW and TRD without conflicting interests or priorities.

The Government will evaluate the adequacy and sufficiency of the manufacturing processes and capability of the Prime and Major Teammates. The Government will evaluate the extent of participation of small businesses, nontraditional defense contractors, and/or nonprofit research institutions. Finally, the Government will assess the implications of the Offeror's asserted technical data rights and how they could impact the Government's ability to operate and sustain T3 Tracking Layer over the life of the constellation.

6.1.5 Factor 5 – Overall On-Orbit Performance Confidence Rating

The Government will assign an Overall On-Orbit Performance Confidence Rating based on the Offeror's proposed maximum liability for the On Orbit Penalty Payback.

Table 6-1 Overall On-Orbit Performance Confidence Rating

Confidence Rating	**Description**
High Confidence	The Offeror proposed at least 10% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.
Substantial Confidence	The Offeror proposed at least 7.50% but less than 10% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.
Satisfactory Confidence	The Offeror proposed at least 5.00% but less than 7.50% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.
Low Confidence	The Offeror proposed less than 5.00% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.

6.1.6 Factor 6 – Price

The Government will evaluate the Offeror's pricing for reasonableness, overall price, and consistency with the profile identified in Table 4-2. Payment Schedule as part of a best value analysis as described in Section 4.4. Elements of the proposed price that may be used in the evaluation include comparison between Offeror proposed prices received in response to the solicitation, comparison of proposed prices with independent Government cost estimates, and analysis of data other than certified cost or pricing data provided by the Offeror. The Government may use data external to the Offeror's proposal such as, but not limited to, field pricing reports, industry information, Government estimates, same or similar DoD contracts, and commercial data when evaluating price reasonableness.

The Government will calculate the total price for the evaluation by adding the total price for all options to the total price for the basic requirement. Evaluation of options will not obligate the Government to exercise the option(s).

6.2 Use of Non-Government Advisors

The Offeror is advised that technical and price data submitted to the Government in response to this solicitation may be released to non-Government advisors that have signed non-disclosure agreements for review and analysis. These non-Government advisors are employed by the following entities:

- AirIn Technologies, Inc
- Alpha 2, Inc.
- Altus LLC
- Amentum Services, Inc. (formerly CENTRA)
- Bryce Space & Technology, LLC
- Custom Analysis LLC
- Emolbi, LLC
- Full Rev LLC
- GoLion Company
- LinQuest, Corporation
- MMAX, Inc.
- MTSI
- PL Consulting, Inc.
- QinetiQ
- Qualis Corporation
- Sigmatch, Inc.
- Space Systems Integration, LLC
- Space Analysis, Inc.
- Strategic Engineering Solutions, LLC
- TEC Solutions, LLC
- The Stacia Group, LLC
- The Aerospace Corporation
- The MITRE Corporation

All non-Government advisors performing this role are expressly prohibited from performing SDA-sponsored technical research and are bound by appropriate nondisclosure agreements. Any objection to release of the Offeror's proposal information to non-Government advisors shall be provided in writing to the Contracting Officer within ten (10) days of the date of issuance (4 April 2025) and shall include a detailed statement of the basis for the objection. The detailed statement shall identify the specific portions of the proposal for which the Offeror objects to disclosure to non-Government advisors.

Advisors from the above entities will continue to support SDA with program management activities throughout the duration of the program. These activities will require ongoing access to technical/price data generated under the agreements as well as the ability to participate in program reviews or technical interchange meetings. Any advisors will be bound by appropriate nondisclosure agreements with SDA. Performers may request copies of the nondisclosure agreements for individual advisors post-award. SDA advisors will not enter into separate NDAs with the performers.

6.3 Number and Scope of Agreements to be Awarded

The Government intends to select up to three (3) Offers, resulting in the total procurement of approximately 54 T3 Tracking Layer SVs and associated ground systems. The Government intends to select a mix of MW/MT and MWTD offers. However, the Government reserves the right to make fewer than three (3) awards, more than three (3) awards, partial awards, or no awards at all for MW/MT SVs and fewer than three (3) awards, more than three (3) awards, partial awards, or no awards at all for MWTD SVs. The Government intends to award a single award for eighteen (18) SVs to three (3) different Offerors, however the Government reserves the right to award more than eighteen (18) to a single Offeror based upon best value.

6.4 Negotiations and Award

As soon as the evaluation of proposals is complete, the Offerors will be notified that (1) the proposal has been selected for funding, subject to OT Agreement negotiations, or (2) the proposal has not been selected for funding. If the Government cannot come to terms with an Offeror selected for funding, it reserves the right to exclude that Offeror and to open negotiations with other Offerors that were not initially selected. Alternatively, the Government may defer a funding decision and first establish a competitive range. If a competitive range is established the Government will conduct negotiations with each offeror included in the competitive range and request final proposal updates. The final proposal updates will be reevaluated in accordance with the criteria set forth in this solicitation. All notifications will be sent via Electronic Mail to the Technical and Administrative POCs identified on the proposal coversheet.

Work will commence after the parties execute the OT agreement. It is SDA's objective to announce the selection of the awardees on or about September/October 2025.

6.5 Other Administrative Information

Assuming the receipt of one or more suitable proposals, the Government reserves the right to select for negotiation all, one, or none of the proposals received in response to this solicitation.

Attachments

XX. Past Performance Questionnaire (PPQ)

XX. Résumé Template

XX. Personnel & Facilities Template

XX. Personnel Matrix